PERFORMING RIGS

AutoPilot Firmware Validation Report

Document	AP-VVR-002
Revision	Rev A
Product	AutoPilot
Hardware Revision	Current Production Hardware
Firmware Revision	Trigger Mitigation Beta (July 2026)
Validation Date	July 22, 2026
Prepared By	Nathan Destro, Performing Rigs
Location	Off-Site Commercial Studio

Validation Summary

Validation Status	OPEN
Functional Result	CONDITIONAL PASS

Validation Status — Validation remains open pending identification, corrective action, and successful revalidation of the intermittent false-trigger condition. Unexpected machine motion is unacceptable for production use. The inability to reproduce the behavior after a power cycle does not reduce its severity or close the issue.

Functional Result — Relay Contact mode and 12/24V Trigger mode successfully recognized repeated intentional trigger events following reconnection, with no missed trigger events observed. Functional objectives under the validated intentional-test conditions were met; this does not constitute release approval.

1. Executive Summary

Additional validation of the AutoPilot Trigger Input was performed using a Mark Roberts Motion Control (MRMC) robot in a commercial production environment.

During the validation session, AutoPilot initially failed to respond to several expected trigger sources, including MRMC relay contacts, a 9V signal routed through a relay, and an independent 24V source. Following reconnection of the system, these failures could not be reproduced, and both Relay Contact mode and 12/24V Trigger mode subsequently demonstrated reliable operation through repeated intentional trigger testing.

However, AutoPilot also exhibited unintended trigger events without any intentional input early in the validation session. Although this behavior disappeared after approximately ten minutes of operation and could not be reproduced following a complete power cycle, unexpected machine motion represents a critical safety concern. Because the root cause remains unknown, the issue cannot be considered resolved despite the inability to reproduce it.

The objective of this report is to document both the successful intentional-trigger validation results and the remaining open engineering and safety concerns.

2. Evidence Summary

Validation Item	Result

Relay Contact Mode	PASS
12/24V Trigger Mode	PASS
Intentional Trigger Misses	NONE
False Triggering	OBSERVED
False Triggering After Power Cycle	NOT OBSERVED

Relay Contact Mode

- MRMC dry-contact relay successfully triggered AutoPilot.
- Approximately 10–15 intentional trigger events were performed.
- Every trigger was recognized successfully.
- No missed trigger events were observed.

12/24V Trigger Mode

- Direct 9V trigger signal successfully activated AutoPilot while configured for 12/24V Trigger Mode.
- Approximately 10–12 intentional trigger events were performed.
- Every trigger was recognized successfully.
- No missed trigger events were observed.

System Stability After Power Cycle

Following a complete power cycle:

- No additional false triggers were observed.
- No unintended trigger events occurred while the MRMC robot performed unrelated motion.
- No false trigger events were observed during continued 12/24V testing.

These observations demonstrate that the previously observed intermittent false-trigger behavior could not be reproduced under the subsequent validation conditions. Non-reproduction does not close F-002.

3. Findings

F-001 — Initial Trigger Failures Were Not Repeatable

Early in testing, AutoPilot did not respond to several expected trigger sources (MRMC dry contact; 9V routed through a relay to dry contact; independent 24V). A Dragonframe DMC-32 positive control confirmed the studio trigger plant could fire other equipment.

Following reconnection of the system, these failures could not be reproduced despite repeated testing.

Because the initial failure mechanism remains unknown, additional investigation is recommended. EMS: AP-010 (closed as not reproduced; chronology retained).

Severity: Medium

Status: Open Investigation

F-002 — Critical Intermittent False Triggering

Early in the validation session, AutoPilot initiated stored moves without any intentional trigger input.

The unintended trigger events occurred approximately every thirty seconds following an early system reset.

The behavior gradually ceased after approximately ten minutes of operation and could not be reproduced following a complete power cycle.

Although the condition became non-repeatable, unexpected machine motion represents a critical safety concern. The inability to reproduce the issue does not reduce its severity or eliminate the need for root-cause analysis. AutoPilot simply cannot false-trigger in production use.

Engineering Impact

- Unexpected machine motion presents an unacceptable operating condition.
- The root cause has not yet been identified (firmware, hardware, electrical noise, initialization, or other).
- The issue should remain classified as unresolved until corrective action has been verified through additional validation.

EMS: AP-001

Severity: Critical

Status: OPEN — Critical Safety Issue

F-003 — Reliable Trigger Recognition Confirmed

Following reconnection and power cycling:

- Relay Contact mode operated reliably.
- 12/24V Trigger mode operated reliably.
- Approximately 20–27 consecutive intentional trigger events were successfully recognized.
- No missed trigger events were observed.

These results demonstrate that the Trigger Input is capable of reliable operation under the tested intentional-trigger conditions.

Severity: Info

Status: Validation Passed

F-004 — 9V Signal Accepted in Both Trigger Modes

A direct 9V trigger signal successfully activated AutoPilot while configured for:

- Relay Contact Mode
- 12/24V Trigger Mode

This behavior was repeatable throughout testing.

Clarification is requested regarding whether this behavior is intentional or represents an unintended characteristic of the Trigger Input circuitry.

Severity: Low

Status: Awaiting Design Clarification

F-005 — Diagnostic Display Limitation

During successful trigger events, the INFO page immediately transitions to the execution screen.

Because of this behavior, trigger voltage and diagnostic information cannot be observed at the moment a successful trigger is accepted.

This limited the ability to capture trigger diagnostic information during validation and to distinguish “signal not seen” from “signal seen but not executed” on future sessions.

EMS: AP-007 / AP-008

Severity: Low

Status: Enhancement Opportunity

4. Questions for Noxon

Q-001 Can the initial trigger failures be explained, given that they could not be reproduced after reconnecting the system?

Q-002 What mechanism could explain the intermittent false-trigger behavior observed early in the validation session (approximately every 30 seconds after reset, self-clearing after ~10 minutes, not reproduced after power cycle)?

Q-003 Is operation from a direct 9V signal intentionally supported in both Relay Contact mode and 12/24V Trigger mode?

Q-004 Would it be possible for the INFO page to retain (latch) the most recent trigger diagnostics after a successful trigger event for troubleshooting purposes?

5. Recommendations

R-001 The intermittent false-trigger condition (F-002 / AP-001) should remain an active critical safety investigation until its root cause has been identified and corrective action has been verified. Do not treat non-reproduction as closure.

R-002 Clarify the intended trigger-voltage thresholds and operating characteristics for each Trigger Input mode (including F-004 / 9V cross-mode behavior).

R-003 Consider implementing a diagnostic mode that preserves trigger voltage and trigger-state information immediately after a successful trigger event (F-005 / AP-007 / AP-008).

6. Remaining Validation

The following validation activities remain planned:

- Extended-duration stability testing
- Additional trigger-source compatibility testing (including known-good 24V path re-test after recovery discipline)
- Long-duration idle monitoring
- Continued validation using MRMC production equipment
- Validation of any future firmware revisions intended to address the intermittent false-trigger condition
- Capture of INFO idle / active readings where workflow allows

Engineering Assessment

This validation session produced two independent conclusions.

First — functional performance. Following reconnection of the system, AutoPilot successfully demonstrated reliable Trigger Input operation in both Relay Contact mode and 12/24V Trigger mode. Approximately 20–27 consecutive intentional trigger events were successfully recognized without any missed triggers.

Second — safety / validation readiness. AutoPilot also exhibited an intermittent false-trigger condition that resulted in unintended machine motion early in the validation session. Although the behavior could not be reproduced following a power cycle, its occurrence represents a critical safety concern because unexpected machine motion is unacceptable during production operation. Likelihood is uncertain; severity remains Critical.

Accordingly:

Dimension	Conclusion
Functional Result	CONDITIONAL PASS
Validation Status	OPEN

Trigger Input functionality has demonstrated reliable performance under repeated intentional testing, but overall Trigger Input validation cannot be considered complete until the source of the intermittent false-trigger behavior has been identified, corrective action implemented, and the system successfully revalidated.

Revision History

Rev	Date	Description
A	2026-07-22	Initial issue — off-site MRMC studio validation (final session conclusions)